

Curriculum Vitae

Tyler Gordon

Postdoctoral Research Associate

University of Arizona, Steward Observatory

Education

2018–2023 **Ph.D. Astronomy and Astrobiology**, University of Washington

2016–2018 **M.S. Astronomy**, University of Washington

2010–2015 **B.S. Physics, Applied Mathematics**, Boise State University

Research

2026– **Postdoctoral Research Associate**, University of Arizona (supervised by Prof. Dániel Apai)

Member of the Pandora Smallsat Mission core science team

Studying the impact of stellar activity on exoplanet transmission spectroscopy.

2023–2026 **Postdoctoral Scholar**, UC Santa Cruz (supervised by Prof. Natalie Batalha)

Characterizing sub-Neptune Exoplanets with JWST transmission spectroscopy

Understanding the sub-Neptune population through atmospheric characterization.

2016–2023 **Graduate Research Assistant**, University of Washington (supervised by Prof. Eric Agol)

Development of differentiable transit models, simulation of JWST observations, and development of a multi-wavelength Gaussian process framework for stellar noise modeling with **Eric Agol**

Measuring stellar rotation in the K2 Sample with **James Davenport**

2012–2016 **Undergraduate Research Assistant**, Boise State University

Simulating sychrotron self-Compton emissions from blazars with Daryl Macomb

Molecular dynamics simulations of soft matter systems with Charles Hanna and David Pink

Grants, Awards, and Honors

- 2026 **(Co-I) JWST Program 11962**
Twinning: An Atmospheric Study of Twin Sub-Neptunes To Investigate Carbon Chemistry, Aerosol Formation, and Photochemistry
- 2025 **(Co-I) JWST Program 12492**
Confirming an 8th planet in the TRAPPIST-1 system
- 2021 **Graduate Student Teaching Award**
- 2020 **(PI) Projectors for the UW Planetarium STF Grant**
Awarded \$75,520 to upgrade planetarium projectors
- 2016–2019 **ARCS Fellowship**

Teaching and Outreach

- Spring 2026 **Instructor, PHYS-182**, University of California, Santa Cruz
Instructor of record for a discipline-specific science communication course for upper-division physics students.
- Fall 2025 **Instructor, ASTR-16**, University of California, Santa Cruz
Instructor of record for a general education astrobiology course.
- 2018–2022 **Planetarium Coordinator**, University of Washington
Recruited and managed a group of 10-15 volunteer presenters and worked in collaboration with undergraduate students to develop age-appropriate planetarium curricula. Initiated efforts to expand the diversity of voices in our planetarium outreach program which eventually led to the department providing an honorarium for a Native American presenter to speak on Indigenous astronomy.

- 2020 – 2022 **Astronomy on Tap Seattle Lead Organizer**
Lead organizer of AoT Seattle as we restarted events after COVID shutdowns. I secured a new venue for the event, recruited co-organizers, scheduled speakers, and was responsible for ensuring that the events ran smoothly.
- 2017 – 2020 **AoT Seattle Co-organizer**
Co-organized Astronomy on Tap Seattle.
- 2016–2017 **Graduate Teaching Assistant**, University of Washington
ASTR 101, ASTR 150, ASTR 270
- 2012–2015 **Undergraduate Teaching Assistant**, Boise State University
PHYS 101, PHYS 102, PHYS 201, PHYS 202

Mentoring

- 2026– **University of Arizona**
Mentored one undergraduate student on a project related to the Pandora smallsat mission.
- 2020 – 2023 **University of Washington**
Mentored two undergraduates in the Pre-Majors in Astronomy Program for one quarter on a project related to stellar rotation
Co-mentored an undergraduate student on a project to measure stellar rotation periods in open clusters
Mentored an undergraduate student on a project to analyze the information content of exomoon/planet mutual transit lightcurves.
- 2024 – **UC Santa Cruz**
Mentored one postbacc and one undergraduate on a project related to the JWST COMPASS program.
Mentored a group of four undergraduate students in the ASTR-9 Intro to Research course on a project related to exomoon detection.
Mentored a group of four undergraduate students in ASTR-9 Intro to Research course on a project related to transit timing and radial velocity exoplanet mass measurements.

Selected Talks

Exploring the Super-Earth and Sub-Neptune Population with JWST (invited), UCSC Institute of Geophysics and Planetary Sciences seminar, Santa Cruz, CA, April, 2026

A Homogeneous Reduction and Analysis of Seven Small Planet Transmission Spectra from the JWST COMPASS Program, Bay Area Exoplanets Meeting #49, San Francisco, CA, November 14, 2025

The Detection and Characterization of Terrestrial Exoplanet and Exomoon Transits in High-Precision Multi-band Photometry, AAS meeting 241, Seattle, WA, January, 2023

Analytic Light Curve for Mutual Transits of Two Bodies Across a Limb-darkened Stars, AAS meeting 240, Pasadena, CA, June, 2022

An Analytic Model for Exomoon Transit Detections with Current and Near-Future Observatories, AbSciCon, Atlanta, GA, May, 2022

The Search for Exomoons (invited), Boise State University, Boise, ID, December, 2021

Measuring Stellar Rotation in the K2 Sample, KITP Online Reunion Conference: Exostar Redux, Santa Barbara, CA, August, 2020

Measuring Stellar Rotation in the K2 Sample, AAS meeting 235, Honolulu, HI, January, 2020

List of Publications

Gordon, Tyler A., Batalha, Natalie M., and 10 colleagues *JWST COMPASS: Insights into the Systematic Noise Properties of NIRSpec/G395H from a Uniform Reanalysis of Seven Transmission Spectra* et al. (2026), The Astronomical Journal, 171, 178.

Gordon, Tyler A. and Agol, Eric *Analytic Light Curve for Mutual Transits of Two Bodies Across a Limb-darkened Star* et al. (2022), The Astronomical Journal, 164, 111.

Gordon, Tyler A., Davenport, James R. A., and 6 colleagues *Stellar Rotation in the K2 Sample: Evidence for Modified Spin-down* et al. (2021), The Astrophysical Journal, 913, 70.

Gordon, Tyler A., Agol, Eric, and Foreman-Mackey, Daniel *A Fast, Two-dimensional Gaussian Process Method Based on Celerite: Applications to Transiting Exoplanet Discovery and Characterization* et al. (2020), The Astronomical Journal, 160, 240.

Batalha, Natasha E., Wallack, Nicole L., and 17 colleagues (including **Gordon, T. A.**) *JWST COMPASS: The 3–5 μm Transmission Spectrum of LTT 1445 A b* et al. (2026), The Astronomical Journal, 172, 52.

Meech, Annabella, Gao, Peter, and 17 colleagues (including **Gordon, T. A.**) *JWST COMPASS: A NIRSpec G395H Transmission Spectrum of Radius Valley Dweller TOI-260 b* et al. (2026), The Astronomical Journal, 171, 274.

Wallack, Nicole L., Gao, Peter, and 18 colleagues (including **Gordon, T. A.**) *JWST COMPASS: NIRSpec/G395H Transmission Observations of the Sub-Neptune HD 15337 c* et al. (2026), The Astronomical Journal, 171, 180.

Hobson-Ritz, Marshall, Birky, Jessica, and 8 colleagues (including **Gordon, T. A.**) *Tidal Synchronization of TESS Eclipsing Binaries* et al. (2025), The Astrophysical Journal, 990, 124.

Teske, Johanna, Batalha, Natasha E., and 17 colleagues (including **Gordon, T. A.**) *JWST COMPASS: NIRSpec/G395H Transmission Observations of TOI-776 c, a 2 $R_{\oplus}$ M Dwarf Planet* et al. (2025), The Astronomical Journal, 169, 249.

Scarsdale, Nicholas, Wogan, Nicholas, and 15 colleagues (including **Gordon, T. A.**) *JWST COMPASS: The 3–5 μm Transmission Spectrum of the Super-Earth L 98-59 c* et al. (2024), The Astronomical Journal, 168, 276.

Foreman-Mackey, Daniel, Luger, Rodrigo, and 14 colleagues (including **Gordon, T. A.**) *exoplanet: Gradient-based probabilistic inference for exoplanet data & other astronomical time series* et al. (2021), The Journal of Open Source Software, 6, 3285.